

Problem Set 2

Due on Canvas at noon on Sept. 30th

Instructions: **No partial credit will be awarded.** All answers must be entered in the Problem Set 2 Quiz on Canvas. Note that it is not possible to edit your responses on Canvas after submitting. **Best practice is to complete the problem set on paper and then transfer solutions to Canvas.** You may collaborate on this problem set with other students in your learning team, but each student must submit his or her own solutions.

Grading: Grading is done on a 15-point scale across all problems in the problem set. **Note that after submitting on canvas, you will see a score, but we will go through your answers manually again to make sure they are graded correctly.** Grades will be finalized a few days after the due date.

Material covered: This problem set covers lectures 5-9.

Problem 1. Tariffs and Import quotas on vegetable fiber

Consider the market for vegetable fiber in the US. The domestic demand for vegetable fiber is given by

$$Q_D(p) = 40 - 2p$$

where Q_D is quantity measured in millions of pounds and P is the price per pound. Vegetable fiber is also produced domestically in the US and the domestic supply curve is given by

$$Q_S(p) = \frac{2}{3}p$$

where Q_S is US output in millions of pounds and p is the domestic price per pound.

- a) Solve for the domestic equilibrium price and quantity of vegetable fiber.
- a. Vegetable fiber trades in a highly competitive world market at a price of \$9 per pound. There is enough availability that the US can import as much as it needs at this price. Do you think US would import vegetable fiber? What would be the amount imported?

☐ Yes, US would import ☐ No, US would not import

Quantity imported: _____

- b. In recent years, the US government imposed a \$3 import tariff. Under this tariff, what do you predict is the domestic price of vegetable fiber?

US domestic price in regime with import tariffs: _____

- c. What would be the total tariff revenue collected by the Government?

Tariff revenue: _____

- d. Without any further computations, can you predict who would be the gainers and losers from such a tariff?

- Relative to free trade, US domestic producer surplus under the import would ...

- ☐ Increase
 - ☐ Decrease
 - ☐ Unclear without more information
 - US domestic consumer surplus (under the tariff vs free trade) would ...
 - ☐ Increase
 - ☐ Decrease
 - ☐ Unclear without more information
 - Total US domestic surplus (under the tariff vs free trade) would ...
 - ☐ Increase
 - ☐ Decrease
 - ☐ Unclear without more information
 - Why did you conclude this about total surplus?
- e. (Challenge question:) Under pressure from trading partners, the US is considering elimination of the tariff. In its place, the proposal is to limit imports into the US to 8 million pounds a year. Under this proposal, what would you predict to be the domestic price of vegetable fiber? What will be the amount produced by domestic producers?

Domestic price: _____

Quantity produced by domestic suppliers: _____

Problem 2. Pet Daycare. Many households either adopted or purchased new fluffy family members during the pandemic. These lovely friends have brought countless joy and love during quarantine. As the society gradually return to normal, business travels, vacations, and even simply commuting to offices have become challenging for new pet parents. Luckily there's an industry for pet-daycare/pet hotel.

Suppose that the weekly cost function for pet daycare (where quantity is in number of pets accommodated a week) is

$$TC(q) = 7.5 + 5q + 0.5q^2$$

(a) What is the marginal cost function for pet daycare?

The industry of pet daycare is arguably quite competitive. Entry is almost free, products are quite undifferentiated, each supplier is small relative to the market.

(b) In this market, what is the supply function of pet daycare **as a function of the market price p for accommodating a pet?** (Specify only the range of prices where the quantity supplied is positive)

(c) Suppose that the demand for pet daycare is given by

$$Q_D(p) = 25 - 2p.$$

What is the (short-run) equilibrium price and quantity? Is it profitable to be a supplier in this industry? What is the profit (if any)?

(d) Knowing that entry is profitable, many others will join the business of pet daycare. What is the long-run equilibrium price? What about quantity?

(e) Being in a competitive industry is not ideal. Suppliers always like to have some market power. One way is to create product differentiation. the *Pooch hotel*, with business locations in Boston ,MA; Chicago, IL; Norwalk, CT; Denver, CO; and Sunnyvale, CA, offers high-quality pet-care services, including presidential suites, webcams for 24/7 monitoring; pet spa; and personal playtime. This captures a new market for high-quality pet daycare. Suppose that the cost function is

$$TC(q) = 15 + 8q + 0.8q^2$$

and that the demand for high – quality pet daycare is

$$Q_D(p) = 250 - 5p.$$

What is the optimal monopoly price for the *Pooch hotel* ? What is the profit? Do you expect this market to become competitive in the long run? Why and why not?

Problem 3. Hair salon versioning. A local hair salon in New Haven is re-designing its business plan. The owner has discovered that they can make more money by offering two packages: “Deluxe” and “Basic”. There are two types of customers: Type A customers usually only need to get haircuts, while Type B customers would enjoy other services (wash, conditioning, coloring, ironing, etc.). Suppose that there are 4 type A customers for each type B customer. The reservations values for each type and marginal costs for each version are as follows:

Reservation Values:	Type A (4)	Type B (1)
Basic (MC=20)	\$30	\$60
Deluxe (MC=50)	\$60	\$90

The optimal menu has the form of: Get the basic package for p_B dollars, or get the deluxe package for p_D dollars.

- What are the constraints that will bind under this menu?
- Use the answers in part a), find the prices p_B and p_D .
- What is the hair salon’s profit by offering this menu?
- What is the hair salon’s profit if it only serves type B by offering the deluxe package only?